SECI1143 PROBABILITY AND STATISTICAL DATA ANALYSIS

SEMESTER 2 (2023/2024)

SECTION 03

PROJECT 1

Analyzing Sleep Patterns and Factors Influencing Quality of Sleep

NAME	MATRIC NO.
YASMIN BATRISYIA BINTI ZAHIRUDDIN	A23CS0201
NAJMA SHAKIRAH BINTI SHAHRULZAMAN	A23CS0140
UTHAYADARSHNI A/P PRAKASH	A23CS5047
AIN NURNABILA BINTI MOHD AZHAR	A23CS0207

LECTURER'S NAME: DR NOR HAIZAN BT MOHAMED RADZI

DATE OF SUBMISSION: 4th MAY 2024

Table of Contents

1.0 Introduction

Our study focuses on analyzing the complex interactions that exist between sleep patterns and the numerous variables that influence the quality of sleep that each person gets. We are curious about this topic because we know how crucial sleep is to preserving general health and cognitive function. As students, we are fascinated by the complex interplay between physiological processes, behavioral, psychological, and environmental factors, and sleep. It is crucial to comprehend these factors to create focused therapies that will improve sleep health and quality of life.

As we begin this inquiry, we expect to come across a wide range of data that reflects the complicated interactions of biological tendencies, lifestyle choices, and society norms. We anticipate seeing trends that show how daily routines, stress levels, screen usage, and the sleep environment affect the quantity, quality, and continuity of sleep. Through close examination of these patterns, we hope to identify both general patterns and individual variances, clarifying the complex ways in which various factors influence sleep-related outcomes.

The main objective of our work is to improve our knowledge of sleep dynamics and provide guidance for evidence-based sleep promotion tactics. We hope to contribute to the creation of interventions that are suited to address particular difficulties people encounter in obtaining restorative and revitalizing sleep by gaining understanding of the variables impacting sleep patterns and quality. With this research, we hope to set the stage for a culture that recognizes sleep as essential to general health and resilience, and where people are empowered to enhance their sleep quality.

2.0 Data Collection

In this project, we used questionnaire via Google Form as a methodology to collect data from UTM students. We successfully gathered 70 respondents in a week. The questionnaire consisted of 3 parts and 20 questions. The first part is where we ask the respondents' biography details, the second is to learn about their sleep patterns, and the third is to find out the factors affecting their sleep.

No.	Parameter/Variable	Level of Measurement	Type of Data
Respondents' Biography			
1	Age	Ratio	Numerical
2	Gender	Nominal	Categorical
3	Year of study	Ordinal	Categorical
4	Faculty	Nominal	Categorical
Sleep Patterns			
5	How important do you think sleep is for your overall health and well-being?	Ordinal	Categorical
6	On a scale of 1 to 5, how would you rate the overall quality of your sleep?	Ordinal	Categorical
7	How many hours of sleep do you typically get on weekdays? E.g. 8 hours	Interval	Numerical
8	Do you have a consistent sleep schedule?	Nominal	Categorical
9	How often do you experience difficulty falling asleep?	Ordinal	Categorical
10	How often do you feel refreshed upon waking up in the morning?	Ordinal	Categorical
11	How many times did you experience sleep disturbances (e.g., snoring, restlessness, nightmares)?	Ordinal	Categorical
12	How many minutes of nap do you take in a day?	Ratio	Numerical
13	How long does it take for you to fall asleep during bedtime? Provide in minutes	Ratio	Numerical
Factors Affecting Sleep			
14	What factors influence the time you go to bed and wake up?	Nominal	Categorical
15	Are there any specific activities or habits you engage in before bed?	Nominal	Categorical
16	Have you noticed any patterns or triggers for vivid dreams or nightmares?	Nominal	Categorical

17	Do you use any electronic devices before bed?	Nominal	Categorical
18	Have you tried relaxation techniques or sleep aids to improve sleep quality (peaceful music, using diffuser/scents)?	Nominal	Categorical
19	How many caffeinated beverages (coffee, tea, energy drinks) do you consume per day?	Ratio	Numerical
20	How many days per week do you engage in physical exercise for at least 30 minutes?	Ratio	Numerical

3.0 Data Analysis

3.1 Respondents biography

3.1.1. Respondent's Age

Age	Frequency
19	6
20	44
21	4
22	7
23	7
24	2
TOTAL	**70**

Table 3.1.1 Frequency Distribution Age of Respondent

Age of each respondent

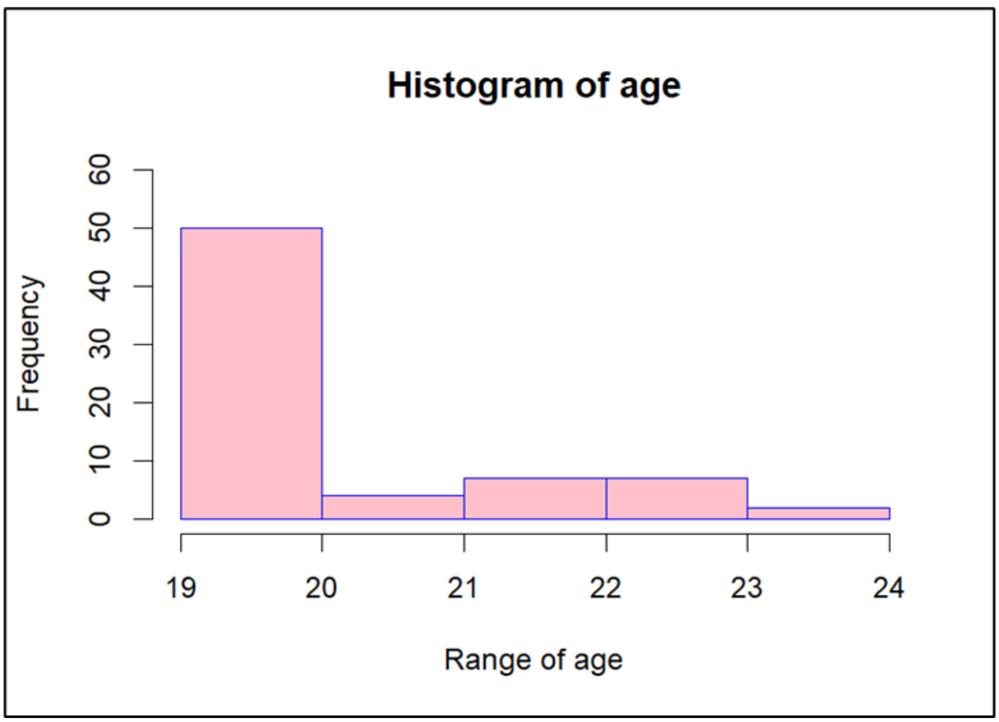

Figure 3.1.1 Histogram Age of Respondent

Measure of Central Tendency	Value
Mean	20.58571
Median	20

Mode	20

Table 3.1.1 Measure of central tendency respondents' age

From table 3.1.1.1 and Figure, the age of distribution of respondents that have answered our questionnaire are around the range of 19 to 24. Since this research is focused on UTM students, most of the respondents are aged 20, with the frequency of 44. Students of age 22 and 23 come close with the frequency of 7, and there is frequency of 6 for students who age 19. There are also respondents who age 21 with the frequency of 4, and respondents who age 24 with frequency of 2. Additionally, it can be seen from table 3.1.1.2 that the data we gathered are mostly from respondents aged 20, as our media and mode both are 20, and mean being 20.58571.

3.1.2 Respondents Gender

Gender	Frequency	Percentage
Male	19	27.14
Female	51	72.86
TOTAL	**70**	**100**

Table 3.1.2 Frequency and percentage of respondents' gender

Gender of Respondent

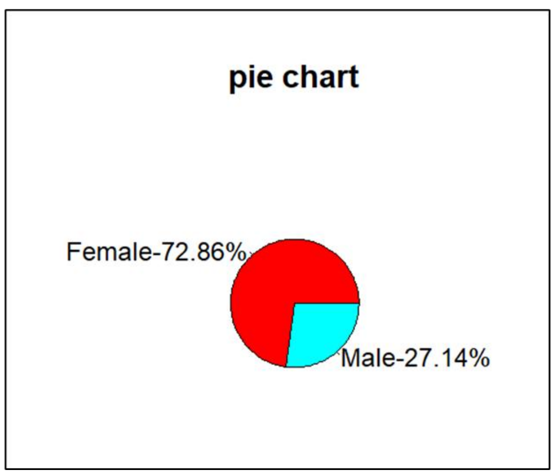

Figure 3.1.2 Pie chart of respondents' gender

As shown in Table 3.1.2.1 and Figure 3.1.2.1, are the frequency table and pie chart of respondents' gender. The frequency of female is larger than male which are female have a frequency of 51, while male have frequency of 19. As a result, female hold more proportion which are 72.86%, and male 27.14%. There are 70 respondents taking part in this questionnaire. In addition, the pie chart of respondent gender illustrates that red means female and blue means male.

3.1.3 Respondents year of study

Year	**Frequency**
1	49
2	16
3	2
4	2
other	1
TOTAL	**70**

Table 3.1.3 Frequency table of respondents' year of study

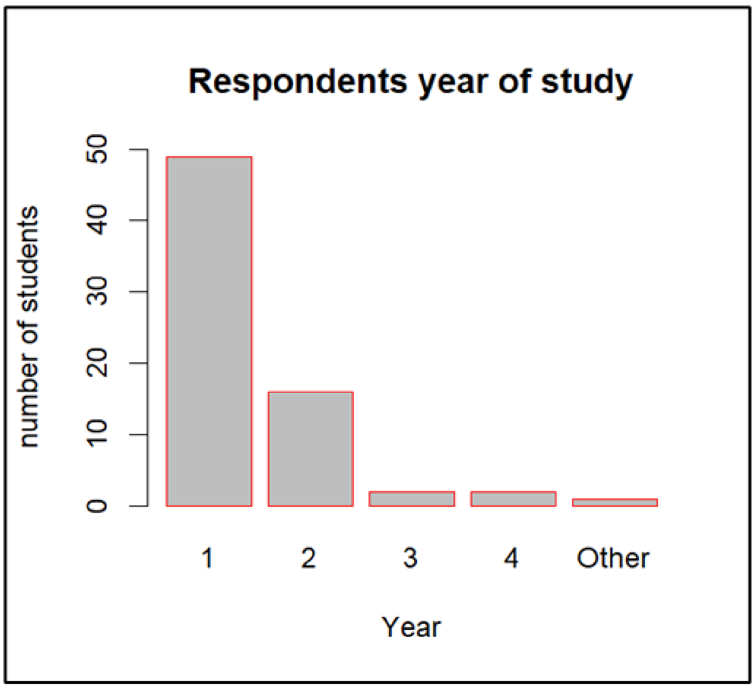

Figure 3.1.3 Bar chart of respondents' year of study

As stated in table 3.1.3.1 and figure 3.1.3.1, the respondents' year of study are from year 1, year 2, year 3, year 4, and other means they are postgraduate students. Mostly our respondents are from year 1 with the frequency of 49. Comes second, year 2 with the frequency of 16. Both year 3 and year 4 have the same frequency, which is 2. And lastly, there is only one respondent who is from postgraduate students. We can conclude that 49 out of 70 respondents are year 1, as well as the mode of respondents' year of study.

3.1.4 Respondents Faculty

Faculty	Frequency	Percentage (%)
Faculty of Build Environment and Surveying (FABU)	2	2.86
Faculty of Chemical & Energy Engineering (FKT)	1	1.43
Faculty of Civil Engineering (FKA)	1	1.43
Faculty of Computing (FC)	45	64.29
Faculty of Electrical Engineering (FKE)	4	5.71
Faculty of Management (FM)	2	2.86
Faculty of Mechanical Engineering (FKM)	9	12.89
Faculty of Science (FS)	3	4.29
Faculty of Social Sciences & Humanities (FSSH)	2	2.86
Malaysia-Japan International Institute of Technology (MJIIT)	1	1.43
TOTAL	**70**	**100**

Table 3.1.4 Frequency and percentage of respondents' year of study

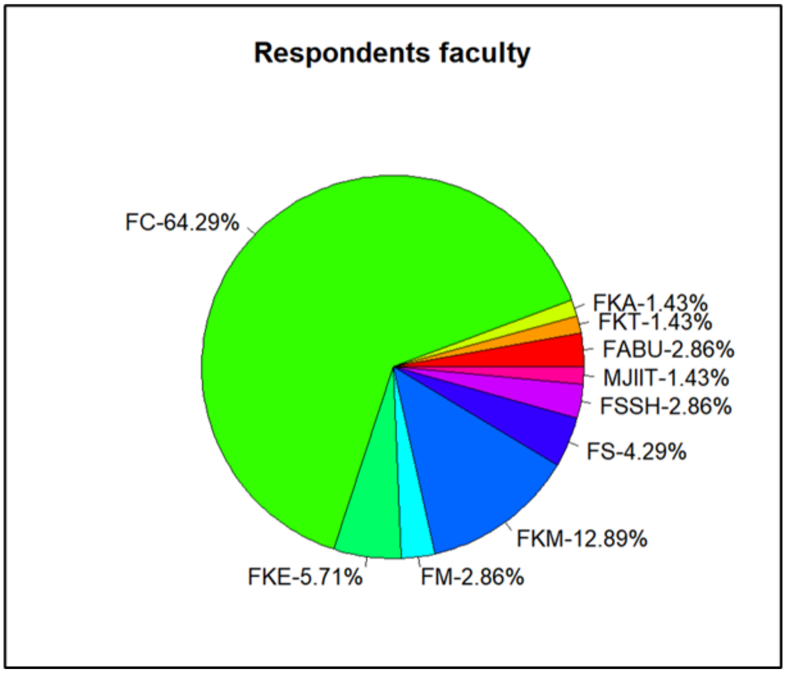

Figure 3.1.4 Pie chart of respondents' faculty

As indicated in Table 3.1.4 and Figure 3.1.4 are the frequency table and pie chart of respondents' faculty. The biggest frequency holds 49 respondents which are from the Faculty of Computing (FC). Comes second is the Faculty of Mechanical Engineering (FKM) with the frequency of 9. Comes third is the Faculty of Electrical Engineering (FKE) with the frequency of 4. And then there are 3 frequencies of faculty of science. Faculty of Build Environment and Surveying (FABU), Faculty of Management (FM), and Faculty of Social Sciences & Humanities (FSSH) have the same frequency, which is 2, and lastly, the Malaysia-Japan International Institute of Technology (MJIIT) holds the frequency of 1 only. There are 70 respondents who took part in this questionnaire.

3.2 Respondents sleep pattern

3.2.1 Scale of Importance of Sleep for Health and Overall Health and Wellbeing

Likert Scale	Frequency	Proportion	Percentage
So not important	0	0.0000	0%
Not important	0	0.0000	0%
Moderate	0	0.0000	0%
Important	12	0.1714	17.14%
Very important	58	0.8286	82.86%
Total	70	1	100%

Table 3.2.1 Frequency, Proportion and Percentage of Respondents Scale of Importance of Sleep for Overall Health and Wellbeing

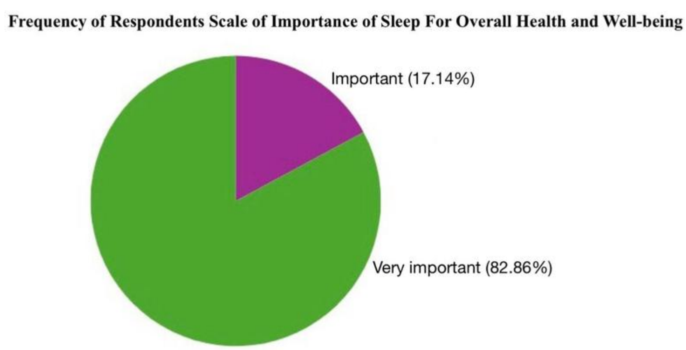

Figure 3.2.1 Pie Chart of Frequency of Respondents Scale of Importance of Sleep For Overall Health and Well-being

For what has been obtained, there is no response for the scale 1-3 (Rarely-Moderately) which indicate 0% on the piechart. However, 17.14% of the respondents think that sleep is important for our overall health and well-being meanwhile 82.86% of the respondents think that the importance of sleep is very crucial for our health as they voted for very important.

3.2.2 Scale of Quality of Sleep

Likert Scale	Frequency	Proportion	Percentage
1	1	0.0143	1.43%
2	15	0.2143	21.43%
3	34	0.4857	48.57%
4	13	0.1857	18.57%
5	7	0.1000	10.00%
Total	70	1	100%

Table 3.2.2 Frequency, Proportion and Percentage of Respondents Scale of Quality of Sleep

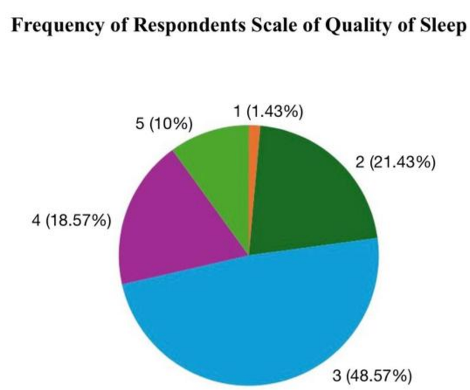

Figure 3.2.2 Pie Chart of Frequency of Respondents Scale of Quality Of Sleep

Based on the data that has been collected, we can see that 1.43% of the respondents rate scale 1 for their overall quality of sleep which refers to bad quality of sleep. Then, 21.43% rated for scale 2 for their quality of sleep. 48.57% rated for scale 3, 18.57% rated for scale 4 and the remaining 10% vote for scale 5 for their overall quality of sleep. We can observe that many of the respondents rated for scale 3 which refers to moderate quality of sleep.

3.2.3 Number of Sleep Hours During Weekdays

Time Interval (hours)	Frequency
3 - 4	8
4 - 5	9
5 - 6	13
6 - 7	23
7 - 8	13
8 - 9	4

Table 3.2.3 Frequency, Proportion and Percentage of Respondents Hours of Sleep On Weekdays

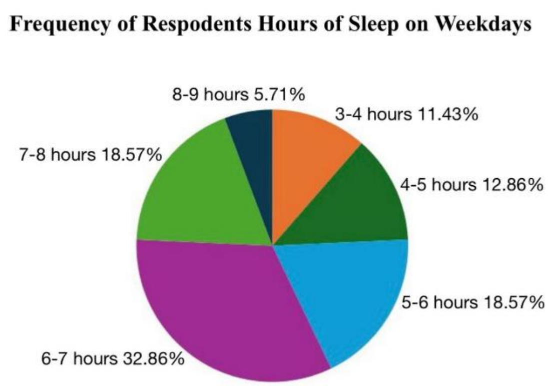

Table 3.2.3 Pie Chart of Respondents Hours of Sleep on Weekdays

Based on the pie chart, there are 11.43% of the respondents sleep for 3 to 4 hours during weekdays which indicates a bad sleeping habit. Besides, we have 12.86% respondents voted for 4 to 5 hours of sleep, 18.57% respondents voted for 5 to 6 hours of sleep, 32.86% respondents for 6 to 7 hours of sleep which stands for majority of the respondents, 18.57% respondents sleep for 7 to 8 hours and only 5.71% of the respondents sleep for 8 to 9 hours on weekdays.

3.2.4 Sleep Schedule Consistency

YES / NO	FREQUENCY
YES	18
NO	52

Table 3.2.4 Frequency, Proportion and Percentage of Respondents Sleep Schedule Consistency

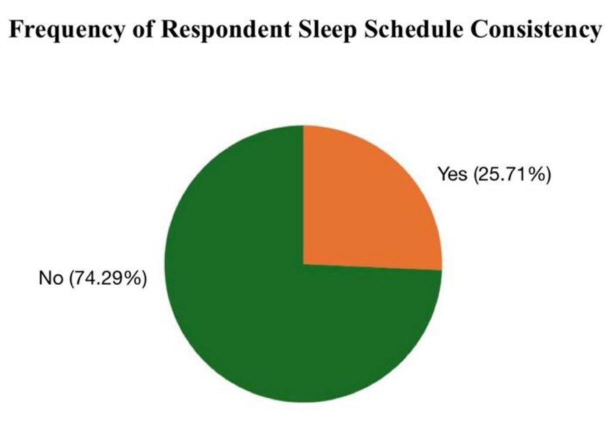

Figure 3.2.4 Pie Chart of Frequency of Respondents Sleep Schedule Consistency

Based on the pie chart above, majority (74.29%) of the respondents have inconsistent sleep schedule. Not getting enough sleep can really throw a wrench into your day. It can make you feel groggy, unfocused, and irritable. Plus, it can affect your memory, decision-making, and overall productivity. Chronic sleep deprivation can even lead to more serious health issues over time. It's important to prioritize good sleep hygiene and seek help if you're consistently struggling to get a good night's rest. However, the remaining (25.71%) respondents are able to make sure they have a consistent sleep schedule. Maintaining a consistent sleep schedule offers a myriad of benefits for both body and mind. By going to bed and waking up at the same time every day, you synchronize your body's internal clock, leading to better sleep quality and enhanced restorative processes during the night. This translates into improved mood, sharper cognitive function, and increased energy levels throughout the day.

3.2.5 Frequency of Respondents Having Sleep Difficulty Falling Asleep

Likert Scale	Frequency	Proportion	Percentage
Rarely	12	0.171428571	17.14%
Sometimes	11	0.157142857	15.71%
Moderately	23	0.328571429	32.86%
Often	13	0.185714286	18.57%
Every Night	11	0.157142857	15.71%
Total	70	1	100%

Table 3.2.5 Frequency, Proportion and Percentage of Respondents Having Sleep Difficulty Falling Asleep

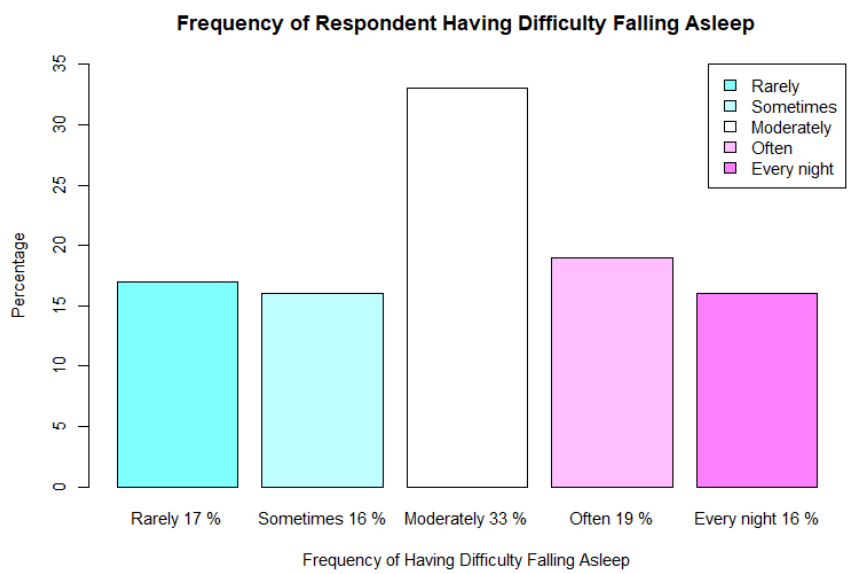

Figure 3.2.5 Bar Chart of Frequency of Respondents Having Sleep Difficulty Falling Asleep

From Figure 3.2.5, The highest percentage among all categories is that 33% of respondents, which is 23 respondents answered that they moderately have trouble falling asleep followed by Often with 19%, Rarely with 17% and Sometimes and Every Night with 16%. This might be alarming as more than half (68%) of the respondents answered having moderate to always having difficulty sleeping. This suggests that UTM students may have a higher prevalence of sleep issues from academic pressure and irregular sleep pattern which ultimately cause difficulty to fall asleep.

3.2.6 Frequency of Respondents Waking Up Feeling Refreshed

Likert Scale	Frequency	Proportion	Percentage
Rarely	5	0.071428571	7.14%
Sometimes	17	0.242857143	24.29%
Moderately	27	0.385714286	38.57%
Often	15	0.214285714	21.43%
Every Night	6	0.085714286	8.57%
Total	70	1	100%

Table 3.2.6 Frequency, Proportion and Percentage of Respondents Waking Up Feeling Refreshed

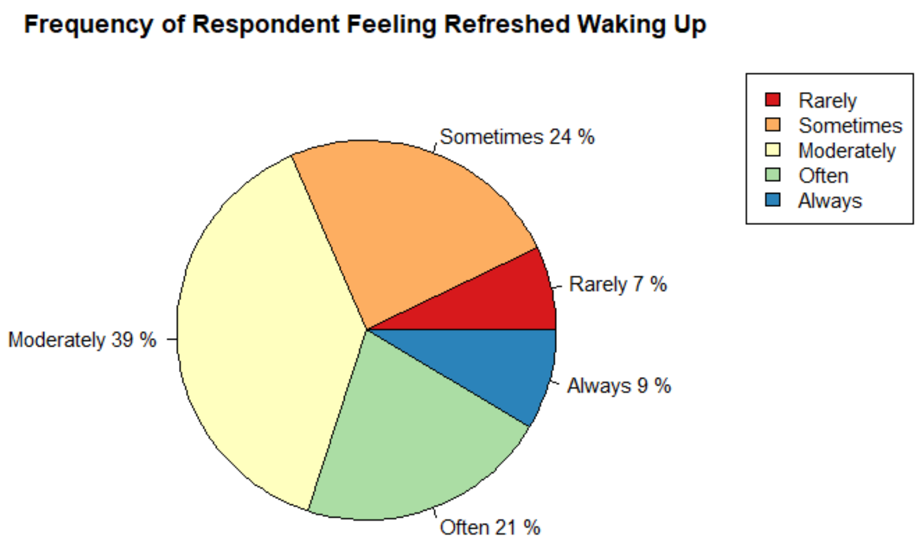

Figure 3.2.6 Pie Chart of Frequency of Respondents Waking Up Feeling Refreshed

Figure 3.2.6 shows that most respondents (39%) answered that they moderately feel refreshed as they wake up. Combined with often (21%) and always (9%) waking up feeling refreshed, 69% of respondents feel a sense of refreshment when they wake up. However, respondents that answered only feeling refreshed sometimes is 12% and rarely feeling refreshed is 8%. This concludes that there may be room for improvement for overall sleep quality and lifestyle habits to feel refreshed when waking up and avoid feeling groggy and tired.

3.2.7 Frequency of Respondents Experiencing Sleep Disturbances

Likert Scale	Frequency	Proportion	Percentage
Rarely	29	0.414285714	41.43%
Occasionally	26	0.371428571	37.14%
Frequently	11	0.157142857	15.71%
Every Night	4	0.057142857	5.71%
Total	70	1	100%

Table 3.2.7 Frequency, Proportion and Percentage of Respondents Experiencing Sleep Disturbances

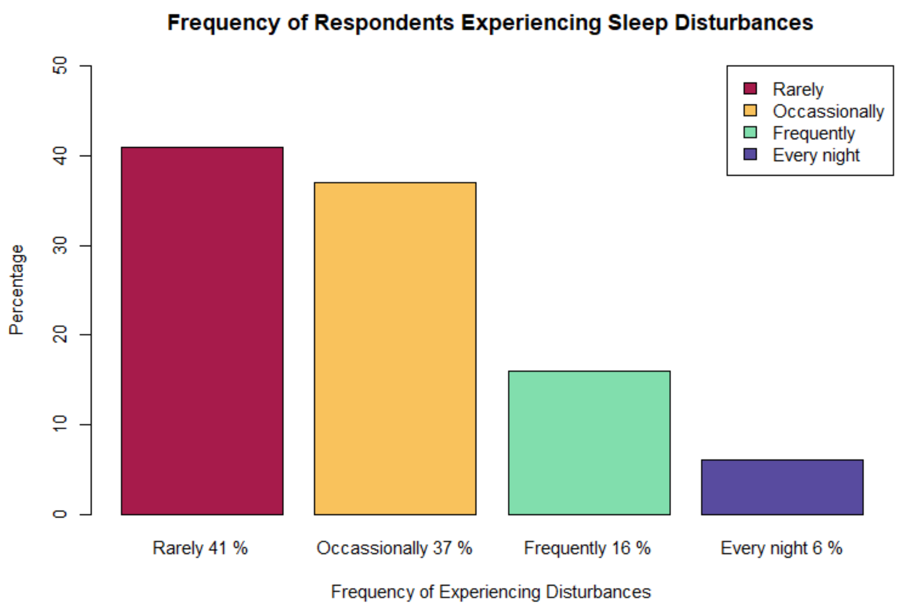

Figure 3.2.7 Bar Chart of Frequency of Respondents Experiencing Sleep Disturbances

The Figure 3.2.7 shows the highest frequency is Rarely with 41%, followed by Occasionally (37%), Frequently (16%) and Every Night (6%). This graph shows that sleep disturbance is rare among most respondents. The data suggest that while a majority of respondents rarely experience sleep disturbances, there are still a significant number of respondents that do and there might be certain factors that are causing it to occur. It is important to address these issues so we can help those who are affected by sleep disturbances.

3.2.8 Relationship Between Napping and Sleep Hours

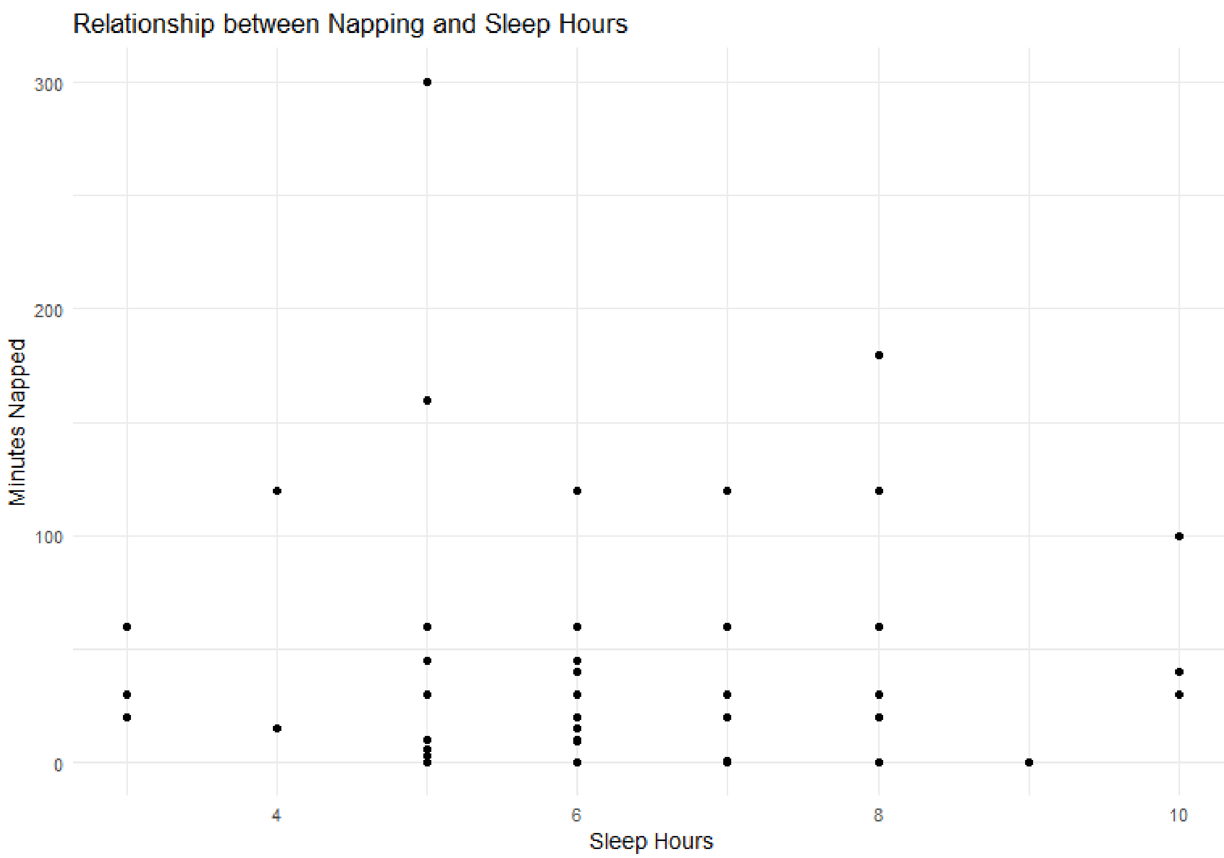

Figure 3.2.8 Scatterplot of Respondents Sleep Hours and Napping Minutes

Figure 3.2.8 shows the relationship between Respondents' sleep hours and napping minutes. The scatter plot shows that as sleep hours increase, napping duration tends to decrease. Most respondents report sleeping for 6 hours with taking a nap from 0 minutes to 120 minutes (about 2 hours). Respondents who get less sleep are more likely to nap during the day suggesting a balance between nighttime sleep and daytime napping.

3.2.9 Time Taken for Respondents to Fall Asleep

Time (minutes)	Frequency
1	1
2	2
3	1
5	8
10	14
15	4
20	5
30	17
40	3
45	3
50	1
60	5
90	1
120	3
150	1
180	1
TOTAL	**70**

Table 3.2.9 is the frequency table of the time taken for the respondents to fall asleep

Quartile	Frequency
0.25	10.0
0.5	25.0
0.75	37.5

Table 3.2.9 shows the frequency of 1st quartile, 2nd quartile, and 3rd quatile

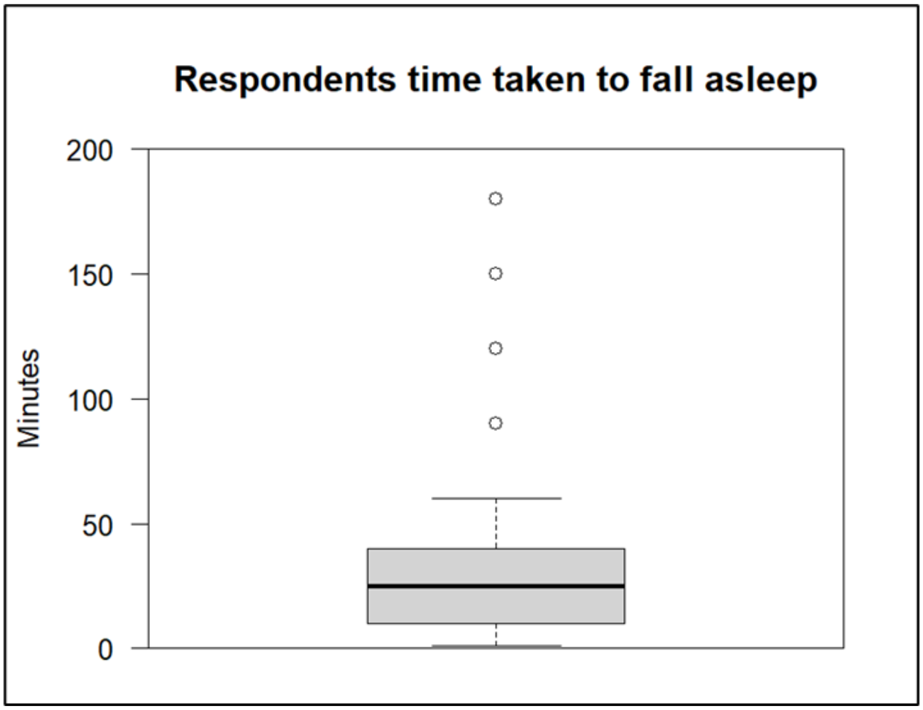

Figure 3.2.9 is the box plot of time taken for respondents to fall asleep

In table 3.2.9 is the frequency of time taken for respondents to fall asleep. The time taken for most of our respondents is 30 minutes with the frequency of 17. Comes second is the time taken for our respondents to fall asleep is 10 minutes, with the frequency of 14. And lastly there is 1 frequency of respondents that took 180 minutes (about 3 hours) to all sleep. We displayed the data into boxplot shown in Figure 3.2.9.

In Figure 3.2.9, we can see that our box plot is slightly skewed to bottom, indicating that most of the respondents would fall asleep in less than 50 minutes. In table 3.2.9, our first quartile lies on 10.0, second quartile lies on 25.0, and third quartile lies on 37.5. There are also 4 outliers exist, with the time of 90, 120, 150, and 180.

3.3 Factors affecting sleep

3.3.1 Factors that Influence the Time Respondents Sleep and Wake Up

Factors	Frequency
All the above	24
Personal preference	2
Social activities & Personal preference	1
Work or study schedule	12
Work or study schedule & Personal preference	15
Work or study schedule & Social activities	13
Work or study schedule, Social activities, Personal preference, All of the above	3
TOTAL	**70**

Table 3.3.1 Frequency of Factors that Influence the Time Respondents Sleep and Wake Up

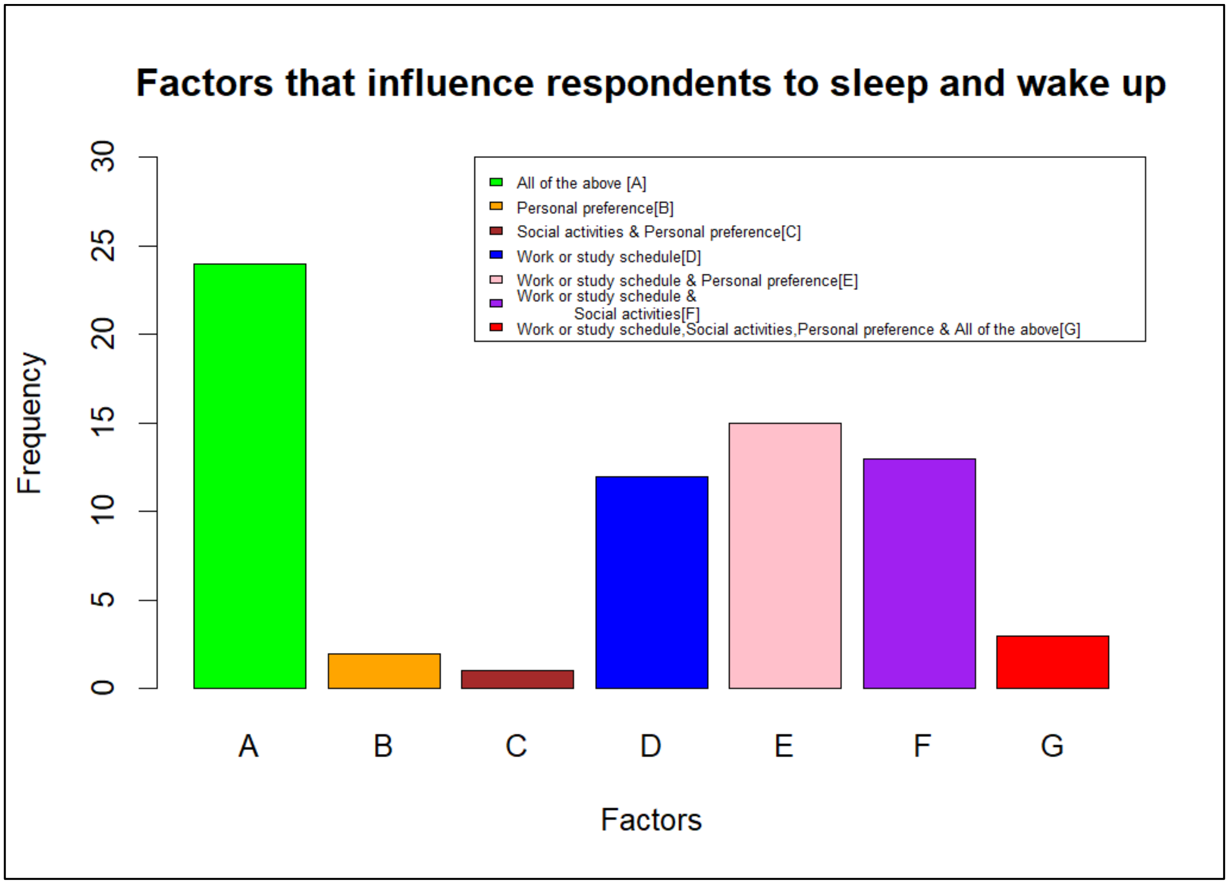

Figure 3.3.1 Bar chart of Factors that Influence the Time Respondents Sleep and Wake Up

From table 3.3.1 and figure 3.3.1, we can see the frequency of each factor that influenced the time respondents to sleep and wake up. All the above choices have the highest frequency of 24, which include Personal preference, Work or study schedule, and social activities. Comes second is with the frequency of 15 which include Work or study schedule & Personal preference. Comes third is with the frequency of 13 which includes Work or study schedule & Social activities. Next is the frequency of 12 which includes Work or study schedule. The third last is with the frequency of 3 which includes Work or study schedule, social activities, Personal preference, All the above. Second last comes with the frequency of 2 which is Personal preference, and lastly with the frequency of 1 is social activities & Personal preference. This means that our sample of population is sleep and wake up for All the above; Personal preference, Work or study schedule, and social activities.

3.3.2 Respondents' activity before sleep

Activity	Frequency	Percentage (%)
Reading	10	14.29
Watching TV or electronic devices	54	77.14
Exercise	0	0
All of the above	6	8.57
TOTAL	**70**	**100**

Table 3.3.2 Frequency and Percentage of Respondents' Activity Before Sleep

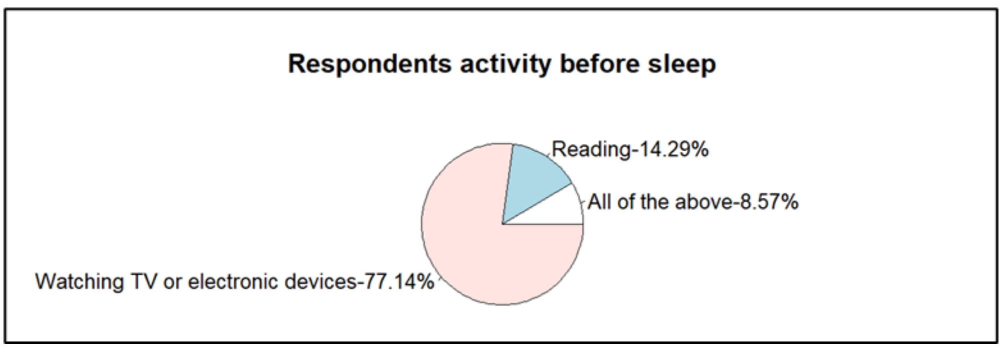

Figure 3.3.2 Pie chart of Respondents activity before sleep

As shows in table 3.3.2 and figure 3.3.2, the majority of the respondents' activity before they went to sleep is by watching TV or electronic devices which has the frequency of 54, and the percentage of it is 77.14%. Comes second is the reading activity with the frequency of 10 and the percentage is 14.29%. Lastly, the choices of all the above; Reading, Watching TV or electronic devices, Exercise has a frequency of 6, and the percentage is 8.57%. Therefore, the pie chart shows the percentage of each activity, where the biggest area is the watching TV or electronic devices activity with 77.14%.

3.3.3 Respondents patterns or triggers for vivid dreams or nightmares

Pattern or Triggers	Frequency
Diet	3
Medication	4
Stress	61
All of the above	2
TOTAL	**70**

Table 3.3.3 Frequency of Respondents patterns or triggers for vivid dreams or nightmares

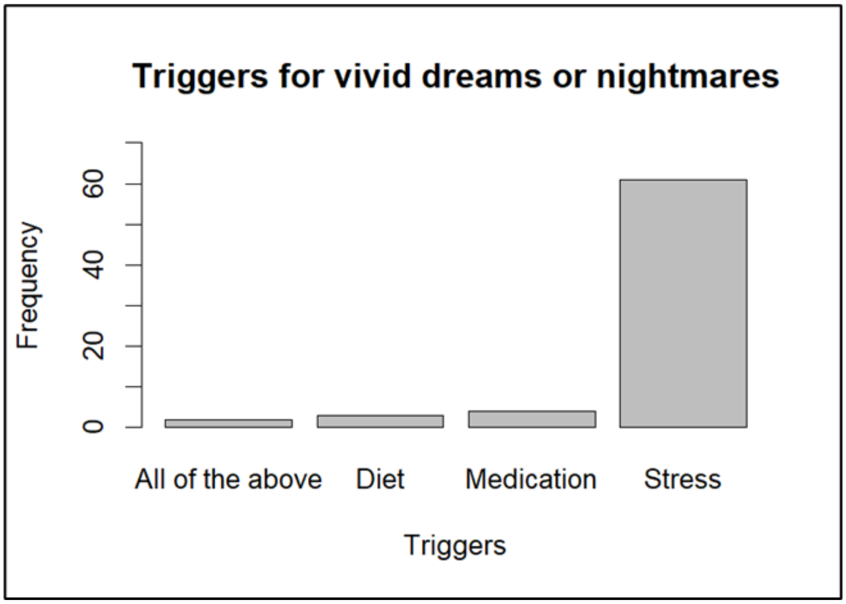

Figure 3.3.3 Bar chart of Respondents patterns or triggers for vivid dreams or nightmares

From table 3.3.3 and figure 3.3.3, we could identify that most of our respondents have patterns or triggers for vivid dreams or nightmares because of stress, where it shows with the frequency of 61. Comes second is because of medication, with the frequency of 4. Next is because of diet, with the frequency of 3. And lastly is because all the above' Diet, medication, and stress with the frequency of 2. Therefore, we could conclude that most of our respondents have vivid dreams or nightmares because of stress.

3.3.4 Respondents Usage of Electronic Devices Before Bed

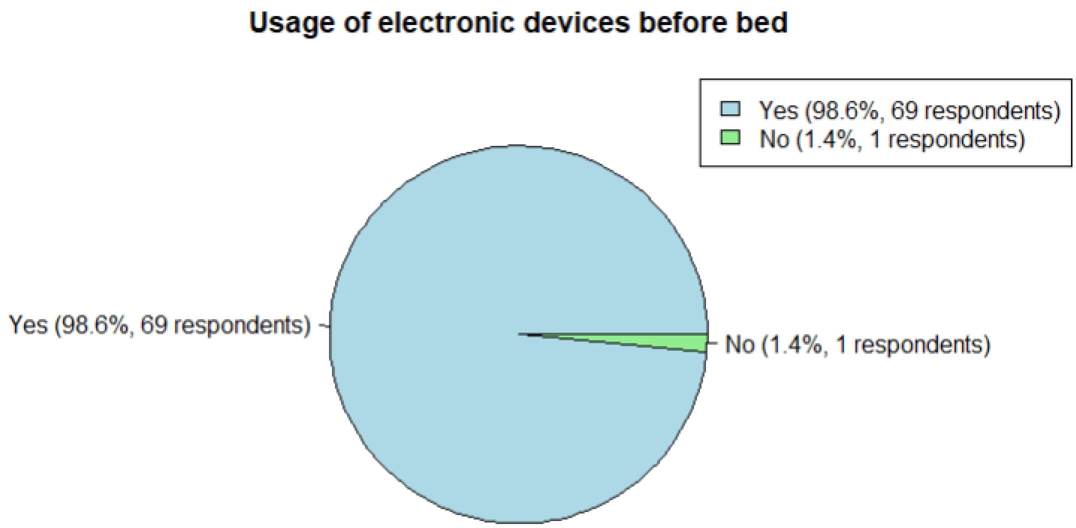

Figure 3.3.4 Usage of Electronic Devices Before Bed

Figure 3.3.4 portrays the usage of electronic devices that the respondents use before their bedtime. Based on this pie chart, 69 respondents, which equals to 98.6% use their electronic device(s) and only 1 respondent which equals to 1.4% does not use his/her electronic device before bedtime.

3.3.5 Respondents Relaxation Techniques to Improve Sleep Quality

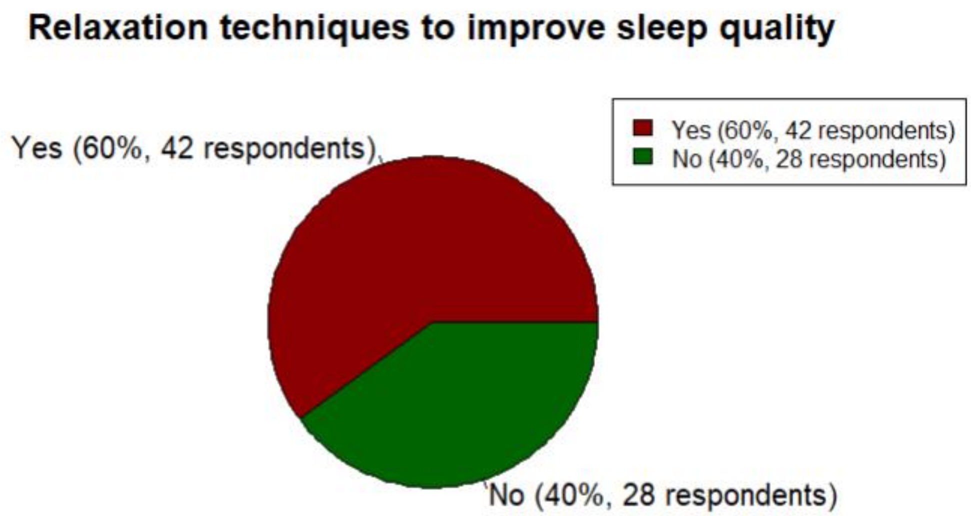

Figure 3.3.5

Figure 3.3.5 shows the relaxation techniques or sleep aids to enhance the respondents sleep quality. The result shows that 42 respondents (60%) use techniques such as peaceful music, diffusers or scents, white noise, herbal remedies and many more to help them sleep. Whereas the remaining 28 respondents (40%) do not use any type of relaxation techniques to sleep.

3.3.6 Number of Caffeinated Beverages Consumed Per Day

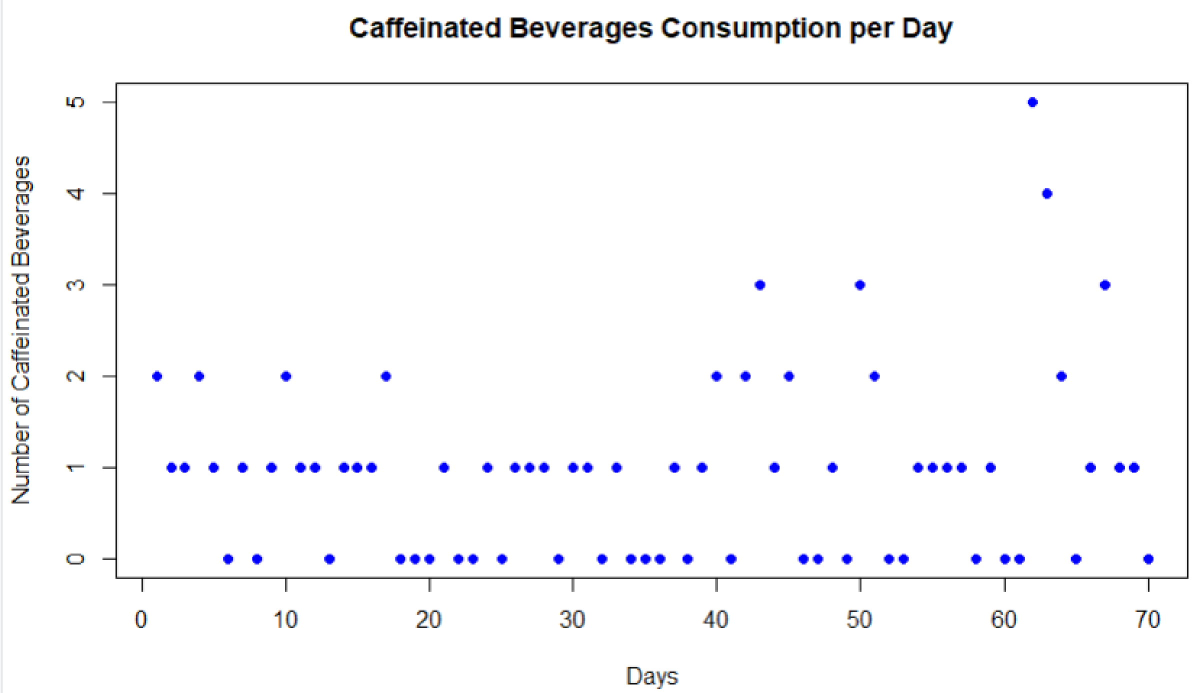

Figure 3.3.6 Scatterplot of Number of Caffeinated Beverages and Days

Figure 3.3.6 illustrates the daily consumption of caffeinated beverages among the respondents. The majority of respondents consume 1 or 2 caffeinated beverages daily according to the data. A significant proportion of respondents indicated that they do not consume any beverages. Moreover, a smaller but statistically significant minority of respondents claimed higher consumption with a few outliers consuming 4 or more caffeinated beverages per day.

3.3.7 Physical Exercise Frequency Per Week for Atleast 30 minutes

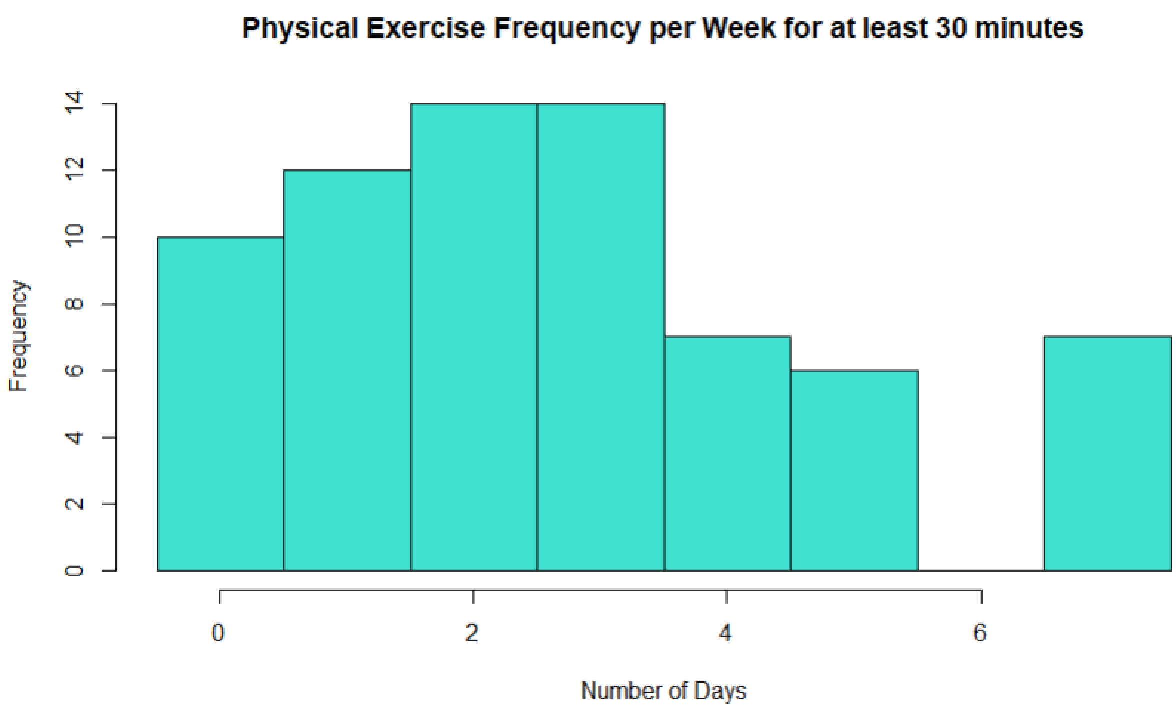

Figure 3.3.7 Histogram of Physical Exercise Frequency per Week for at least 30 minutes

Figure 3.3.7 represents the frequency where the respondents have exercised for at least 30 minutes a week. The histogram above shows that the respondents have a wide variety of exercise habits where most of them work out 1 to 3 days a week. Interestingly, a significant portion of respondents have declared that they did not exercise any day of the week. Conversely, some respondents also show more regular exercise habits and some of them work out 4 to 7 days a week.

4.0 Conclusion

From this survey we have conducted, it can be concluded that most respondents agree that sleeping is important for overall health and well-being. The study's results showed that respondents had a variety of sleep habits and difficulties, with differing evaluations of their quality of sleep. Based on the data, 34 respondents have rated their overall sleep quality around 3 on a scale of 1 to 5. This is due to the factors of work schedules, stress and activities before bed on their sleep patterns. In this survey, it is also shown that many respondents do not have a consistent sleep schedule as they have trouble falling asleep. Moreover, respondents' naps ranged widely from 0 to 300 minutes (about 5 hours), with an average duration of 42.7 minutes each day. There was a notable variation in the napping habits of the chosen population, as some respondents did not have a nap at all in a day, while others slept for extended durations. In addition, 63 respondents also think that the factor affecting their sleep is mostly because of stress. 44 respondents have tried relaxation techniques and sleep aids but many of them use electronic devices right before their bed. Lastly, the respondent's engagement in exercise also varies greatly as most of them exercise 3 to 4 times per week for at least 30 minutes.

Google Form Screenshot

Analyzing Sleep Patterns and Factors Influencing Quality of Sleep

We are 1st year students from Section 03 of SECI1023 Probability and Statistical Data Analysis subject and we are conducting this questionnaire for our project.

Our study aims to explore the sleeping habits and external factors that maybe affecting quality of sleep among UTM students. We thank in advance for answering this questionnaire and highly appreciate your time.

enomototakane345@gmail.com Switch account

* Indicates required question

Email *

Your email

How old are you? *

Your answer

What is your gender? *

- ○ Male
- ○ Female

What is your current year of study? *

- ○ 1
- ○ 2
- ○ 3
- ○ 4
- ○ Other: ________

Which faculty are you from? *

- ○ Faculty of Civil Engineering (FKA)
- ○ Faculty of Mechanical Engineering (FKM)
- ○ Faculty of Chemical & Energy Engineering (FKT)
- ○ Faculty of Electrical Engineering (FKE)
- ○ Faculty of Computing (FC)
- ○ Faculty of Build Environment and Surveying (FABU)
- ○ Faculty of Science (FS)
- ○ Faculty of Social Sciences & Humanities (FSSH)
- ○ Faculty of Management (FM)
- ○ Other: ________

Next

Clear form

Analyzing Sleep Patterns and Factors Influencing Quality of Sleep

enomototakane345@gmail.com Switch account

* Indicates required question

Sleep patterns

How important do you think sleep is for your overall health and well-being? *

	1	2	3	4	5	
Not important	○	○	○	○	○	Very important

On a scale of 1 to 5, how would you rate the overall quality of your sleep? *

	1	2	3	4	5	
Poor	○	○	○	○	○	Great

How many hours of sleep do you typically get on weekdays? Eg. 8 hours *

Your answer

Do you have a consistent sleep schedule? *

○ Yes

○ No

How often do you experience difficulty falling asleep? *

	1	2	3	4	5	
Rarely	○	○	○	○	○	Every night

How often do you feel refreshed upon waking up in the morning? *

	1	2	3	4	5	
Rarely	○	○	○	○	○	Always

How many times did you experience sleep disturbances (e.g., snoring, restlessness, nightmares)? *

○ Rarely

○ Occasionally

○ Frequently

○ Every night

How many minutes of nap do you take in a day? *

Your answer

How long does it take for you to fall asleep during bedtime? Provide in minutes *

Your answer

Back Next Clear form

Analyzing Sleep Patterns and Factors Influencing Quality of Sleep

enomototakane345@gmail.com Switch account

* Indicates required question

Factors affecting sleep

What factors influence the time you go to bed and wake up? *

- [ ] Work or study schedule
- [ ] Social activities
- [ ] Personal preference
- [ ] All of the above

Are there any specific activities or habits you engage before bed? *

- ◯ Exercise
- ◯ Watching TV or electronic devices
- ◯ Reading
- ◯ All of the above

Have you noticed any patterns or triggers for vivid dreams or nightmares? *

- ◯ Stress
- ◯ Diet
- ◯ Medication
- ◯ All of the above

Do you use any electronic devices before bed? *

- ◯ Yes
- ◯ No

Have you tried relaxation techniques or sleep aids to improve sleep quality (peaceful music,using diffuser/scents)? *

- ◯ Yes
- ◯ No

How many caffeinated beverages (coffee, tea, energy drinks) do you consume per day? *

Your answer

How many days per week do you engage in physical exercise for at least 30 minutes? *

Your answer

Back Next Clear form

Analyzing Sleep Patterns and Factors Influencing Quality of Sleep

enomototakane345@gmail.com Switch account

Thank you!

Thanks for taking your time to answer this questionnaire!

Back Submit Clear form